

Ashwin Padaki

✉ apadaki@seas.upenn.edu | [apadaki.github.io](https://github.com/apadaki) | [apadaki](https://www.apadaki.com)

SUMMARY

Computer Science Ph.D. student specializing in fast algorithms for similarity search, clustering, and other high-dimensional problems, with an emphasis on approaches that are theoretically grounded and scalable.

EDUCATION

University of Pennsylvania

Ph.D in Computer and Information Science

Philadelphia, PA

2024 – 2029 (expected)

Advised by Sanjeev Khanna and Erik Waingarten.

Columbia University

B.A. in Computer Science, Mathematics

New York, NY

2020 – 2024

GPA: 4.06/4

PUBLICATIONS

(authors ordered alphabetically by last name)

1. Tian Zhang, **Ashwin Padaki**, Jiaming Liang, Zack Ives, Erik Waingarten. [Prune, Don't Rebuild: Efficiently Tuning \$\alpha\$ -Reachable Graphs for Nearest Neighbor Search.](#) .
(in submission)
2. **Ashwin Padaki**, Styopa Zharkov. Provably Fast Density-Based Clustering in High Dimensions.
(in submission)
3. Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. [Sparse Navigable Graphs for Nearest Neighbor Search: Algorithms and Hardness.](#)
SODA 2026
4. Sanjeev Khanna, **Ashwin Padaki**, Krish Singal, Erik Waingarten. [A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams.](#)
FOCS 2025
5. Henry Fleischmann, Kyrylo Karlov, Karthik C. S., **Ashwin Padaki**, Styopa Zharkov. [Inapproximability of Maximum Diameter Clustering for Few Clusters.](#)
SODA 2025
6. Josh Alman, Yunfeng Guan, **Ashwin Padaki**. [Smaller Low-Depth Circuits for Kronecker Powers.](#)
SODA 2023

IMPLEMENTATION EXPERIENCE

Open-Source Implementations

Python (NumPy, PyTorch), C++, C

- Locality sensitive hashing-based DBSCAN with a fully subquadratic-time guarantee. [dbscan](#)
- Data-driven spectral partitioning for nearest neighbor search (in progress). [data-driven-nns](#)
- Piecewise orthogonal transformation to align embeddings produced by different models. [procrustes-embed](#)

SELECTED WORK EXPERIENCE

Head Teaching Assistant <i>University of Pennsylvania</i>	Aug 2025 – present <i>Philadelphia, PA</i>
Algorithms for Big Data (Spring 2026, Fall 2025).	
Teaching Assistant <i>Columbia University</i>	Sep 2022 – Dec 2023 <i>New York, NY</i>
Cryptography (Fall 2023), Computational Complexity (Spring 2023), Real Analysis (Fall 2022).	
Quantitative Trader Intern <i>Optiver</i>	Jun 2022 – Aug 2022 <i>Chicago, IL</i>

TALKS

1. Sparse Navigable Graphs for Nearest Neighbor Search	
SODA 2026 Conference	Jan 2026
NYC Graduate Student Theory Seminar	Nov 2025
DIMACS/Rutgers Theory Seminar	Nov 2025
CMU Theory Seminar	Sep 2025
2. Inapproximability of Maximum Diameter Clustering for Few Clusters	
SODA 2025 Conference	Jan 2025

SERVICE

Seminar Organizer <i>UPenn Theory Seminar</i>	Sep 2025 – present <i>Philadelphia, PA</i>
Seminar Instructor <i>Columbia Undergraduate Learning Seminar in Theoretical Computer Science</i>	Sep 2023 – Dec 2023 <i>New York, NY</i>

ACHIEVEMENTS

National Science Foundation (NSF) Graduate Research Fellow	2024
Phi Beta Kappa Inductee	2024
Putnam Mathematical Competition, Top 500 Scorer	2022